

IMAGE PROCESSING TECHNIQUES

Extended Class Activities Worksheet | UG Level | Activity 10: Gradient Magnitude Calculation

INPUT DATA

Image Patch (A):

10	10	10
0	0	0
0	0	0

Sobel Kernels:

K_x (Horizontal Gradient)

-1	0	1
-2	0	2
-1	0	1

K_y (Vertical Gradient)

-1	-2	-1
0	0	0
1	2	1

STEP-BY-STEP SOLUTIONS

1. 1. Compute G_x (Horizontal Gradient):

$$G_x = (-1)(10) + (0)(10) + (1)(10) + (-2)(0) + (0)(0) + (2)(0) + (-1)(0) + (0)(0) + (1)(0)$$
$$G_x = -10 + 0 + 10 + 0 + 0 + 0 + 0 + 0 + 0 = 0$$

2. 2. Compute G_y (Vertical Gradient):

$$G_y = (-1)(10) + (-2)(10) + (-1)(10) + (0)(0) + (0)(0) + (0)(0) + (1)(0) + (2)(0) + (1)(0)$$
$$G_y = -10 - 20 - 10 + 0 + 0 + 0 + 0 + 0 + 0 = -40$$

3. 3. Calculate Gradient Magnitude (G):

$$G = \sqrt{G_x^2 + G_y^2}$$
$$G = \sqrt{0^2 + (-40)^2} = \sqrt{1600} = 40$$

4. 4. Identify Edge Type:

Since $G_x = 0$ and $|G_y| = 40$, the intensity change occurs along the vertical axis (row 1 to row 2). Therefore, this represents a **Horizontal Edge**.